Critical Review: "Pivot-Based Approximate Similarity Search for Hospital Encounters"

Bottom-line verdict

Promising topic, but I would not submit the proposal unchanged. The algorithmic focus is strong and well aligned with DS8001, but the current version contains **four approval/marking risks**:

1. **The claim that the proposed mixed-type distance "is a proper metric" is not justified as written.**
2. **MPT is insufficiently defined and appears ambiguously or incorrectly attributed.**
3. **KD-tree, ball-tree, and IVF are not currently specified as apples-to-apples baselines.**
4. **Q3 invokes intrinsic dimensionality without defining or measuring it.**

The proposal also risks **scope collapse**: two distances, four pivot variants, two pivot-selection methods, three baseline families, four values of N, two K's, a three-dimensional parameter grid, ten seeds, and an optional prediction study. DS8001 explicitly prefers a careful investigation of a few well-motivated algorithms over a shallow collection of methods, and its heaviest criteria are Methods and Experiments, at 25% each.

Strict recommendation: approve only after a substantial methodological tightening. As submitted, I would view it as a **good idea with potentially serious correctness and feasibility weaknesses**, not yet a fully defensible experimental plan.

---

1. Highest-priority technical problems

1.1 The metricity claim is overconfident

The proposal says that the mean of range-normalized numeric/ordinal differences and categorical mismatch "is a proper metric." That is **conditionally true, not automatically true**.

It is a metric if all of the following hold:

- Every numeric feature uses a **fixed positive scale** R_j, so
- Every ordinal feature is mapped to a **fixed, ordered numerical scale**, rather than arbitrary integer category IDs.
- Every nominal feature uses the discrete metric: zero for equality and one otherwise.
- Feature weights are **fixed, nonnegative, pair-independent**, with enough positive weights to distinguish different records.

- The denominator is fixed across all pairs.
- Missing values are incorporated through a per-feature metric rather than causing pairwise feature deletion.

Under those conditions, the overall distance is a positive weighted sum of component metrics and is therefore a metric.

Where your formulation can fail

Pairwise omission of missing features is dangerous. Original Gower similarity averages only over comparable variables, and the original paper explicitly notes that missing values may destroy desirable matrix properties. More recent analysis also shows pathological values caused by omitting missing attributes. [jstor.org] [annals-csis.org]

A simple counterexample demonstrates the metric problem:

- $x = (0,\backslash text\{missing\})$
- $y = (0,1)$
- $z = (1,1)$

If missing features are dropped pairwise, then $d(x,y) = 0$, $d(y,z) = 0.5$, and $d(x,z) = 1$. This violates both identity of indiscernibles and the triangle inequality.

"Missing lab result as an explicit category" is defensible **only if it means a fixed state such as** NOT_MEASURED, with zero distance between two such states and a specified distance to every observed state. For categorical lab-result fields such as A1Cresult, that is straightforward. For a genuinely numeric field, define missing–observed distance explicitly—e.g., one after clipping observed normalized values to $[0,1]$.

Other unresolved conditions:

- A zero-range feature makes normalization undefined and must be dropped.
- Ranges must be estimated **from the index/training partition only**, then frozen. Otherwise preprocessing leaks test information.
- Query values outside training ranges require a declared rule, preferably fixed domain bounds or clipping.
- Arbitrary spacing of ordinal codes can distort meaning even while remaining mathematically metric.
- Equal weight per raw column may let the 23 medication fields dominate broader concepts such as diagnosis or demographics.

Defensible rewrite

> "We use a fixed weighted product metric. Numeric and ordinal features use absolute differences after training-set-only scaling; nominal features use 0/1 mismatch. Missingness is represented by fixed feature-specific states rather than pairwise feature deletion. With fixed nonnegative weights and denominator, each component is metric and

> their weighted sum is metric. We will test triangle inequality and identity on sampled triples as an implementation check."

That is far stronger than citing Gower and declaring metricity.

1.2 MPT, MIF, and prefix indexing are conflated

The proposal describes "the MPT scoring refinement, plus a prefix index (Esuli, 2012)." This is too vague to implement or grade.

The literature supports the following distinctions:

- **Permutation-based indexing** represents an object by pivots ordered by distance.
- **MIF/MI-File** stores, for each pivot, an inverted list containing object identifiers and pivot ranks; Spearman footrule is used as a surrogate similarity measure. Truncated forms retain only nearest reference objects. [link.springer.com]
- **PP-Index/permutation-prefix indexing** stores prefixes and places objects with the same prefix together; candidate objects are then verified using true distances. [link.springer.com]
- **MPT—Metric Permutation Table—is a quantized-ranking representation and associated data structure**, not merely an unspecified "scoring refinement." [link.springer.com]

Therefore, the proposal's citation chain is ambiguous. The cited prefix paper may support PP-Index, but **MPT needs its own direct citation and algorithm definition**. The later quantized-ranking paper explicitly presents MPT as its own data structure. [link.springer.com]

Concrete fix

Include four or five lines of pseudocode-level specificity:

1. How the quantized rank is calculated.
2. What an MPT cell stores.
3. How a query generates candidate scores.
4. Which parameter controls quantization.
5. How candidates are selected and exactly reranked.

If you cannot define those precisely on one page, **drop MPT and compare full-rank versus truncated MIF**, or change the project path toward replication of one specific MPT paper.

1.3 Spearman footrule on truncated permutations is underspecified

For full permutations,

The literature supports full-permutation SFD and MIF rank storage. But once only $\tilde{n}$ pivots are retained, some pivots appear in one prefix but not the other. Your proposal does not say: [link.springer.com]

- what rank is assigned to an absent pivot;
- whether scoring is only over intersections;
- whether non-overlap receives a penalty;
- how ties in pivot distance are broken;
- whether objects with zero overlap are ignored or enter with a default score.

These decisions can materially change candidate recall.

Similarly, "recall is limited when a query and true neighbor share no pivot" is **not a universal property of pivot indexing**. It is a candidate-generation failure mode for certain truncated/intersection-based inverted schemes. It does not apply in the same way to full permutation comparison, and may depend on the absent-pivot scoring rule.

Better wording

> "In our truncated inverted implementation, an object with no retained pivot in common with the query cannot receive a positive candidate score unless a fallback mechanism is used; this can exclude a true neighbor before reranking."

That is implementation-specific and testable.

1.4 Complexity statements are incomplete

"Build cost $O(Nn)$ distances" is correct as a **distance-evaluation count**, but it is not a complete build-time analysis.

Also include:

- sorting all pivot distances: $O(Nn\log n)$, unless partial selection is used;
- retaining $\tilde{n}$ entries: selection and rank-encoding costs;
- farthest-first pivot selection cost;
- query pivot distances: n per query;
- accumulator/postings operations;
- candidate selection cost;
- c true distances for reranking.

Likewise, $O(N\tilde{n})$ is an **entry count**, not actual index memory. Each entry may contain object ID, pivot ID or implicit list membership, rank/quantized rank, and container overhead. MPT memory cannot simply inherit the same claim until its representation is defined.

"Exact reranking" must be rewritten:

> "The best c candidates are evaluated with the true distance and exactly ordered within that candidate set; the overall result remains approximate unless all true top-K neighbors entered the candidate set."

---

2. Baseline fairness and suitability

KD-tree and ball-tree

Scikit-learn's KDTree and BallTree do **not support arbitrary callable metrics** in their tree APIs. Therefore they cannot directly consume your custom mixed-type Python distance as implied. [scikit-learn.org], [scikit-learn.org]

They are defensible as exact baselines only in one of these settings:

1. **Dense one-hot Euclidean representation**, using Euclidean tree search.
2. A rigorously derived dense transformation of your mixed distance into a supported weighted Manhattan/Minkowski form.
3. A different library that explicitly supports the same metric.

Scikit-learn also states that sparse input forces NearestNeighbors to use brute force. This matters because one-hot ICD-9 and medical-specialty fields are naturally sparse. Densifying them may materially alter memory comparisons. [scikit-learn.org]

Calling the trees **exact** is appropriate if:

- exact query settings are used;
- their returned neighbors match brute force under the same metric;
- tie handling is aligned.

But exact does not mean efficient: tree pruning may collapse in high-dimensional one-hot spaces. That failure would be a legitimate result. Both classes expose distance-call statistics, which would be useful for machine-independent comparison. [scikit-learn.org], [scikit-learn.org]

IVF

A k-means IVF baseline is meaningful only for the **Euclidean encoding**. K-means coarse quantization is not naturally compatible with the proposed mixed Gower-like metric.

A proper IVFFlat baseline should sweep:

- number of lists;
- number of probed lists;
- training seed;
- candidate vectors scanned.

Official Faiss documentation describes IVF as quantizing a vector to one or more posting lists and searching only those lists; IndexIVFFlat stores raw vectors within the selected lists. [faiss.ai], [faiss.ai]

The Jégou citation is not wrong as background, but it is mismatched if your baseline is plain IVF built with scikit-learn. That paper's central contribution is **product quantization combined with a coarse inverted file**, including residual encoding—IVFADC—not simply uncompressed IVFFlat. [irisa.fr]

Recommended citation choice:

- Cite the Jégou paper only if implementing or discussing IVFADC/PQ.
- Cite official Faiss IVFFlat documentation for the actual baseline.
- If you implement IVF yourself using scikit-learn k-means, say so; it is no longer merely an off-the-shelf baseline.

Apples-to-apples rule

Do **not** compare recall across different ground-truth geometries.

- Mixed metric methods must use mixed-metric brute-force ground truth.
- One-hot Euclidean methods and IVF must use Euclidean ground truth.
- Treat the second encoding as a **sensitivity study**, not as another point on the same recall–cost curve.

---

3. Professor/grader perspective

The project fits DS8001 unusually well: exact versus approximate search is explicitly named as a suitable topic, with recall, memory, query time, and scalability as appropriate dimensions. The self-implementation of brute force, MIF, and MPT could provide a substantive algorithmic component.

However, a strict grader could deduct marks for:

Problem significance

"Supports case-based reasoning" is plausible but not demonstrated. The project actually establishes **algorithmic retrieval performance**, not clinical usefulness. Avoid implying clinical utility unless you evaluate it.

Rewrite: "We use hospital encounters as a challenging mixed-type retrieval benchmark; no clinical-decision claim is made."

Research questions

- **Q1 is too broad:** "which dominates?" ignores interactions among $n,\tilde{}\,n, c$.

- **Q2 does not define "improve":** recall at equal memory, equal postings, equal candidate budget, or equal latency?
- **Q3 is presently unanswerable:** no intrinsic-dimensionality estimator or intervention is specified.

Originality

The originality is primarily **application, controlled replication, and empirical diagnosis**, not a new algorithm. That is acceptable, but say it plainly. Otherwise "our contribution" may sound inflated.

Scope

The guidelines explicitly prefer depth over a large collection of algorithms. Your current plan includes too many moving pieces and several optional ones. A grader may reasonably conclude that the proposal is **insufficiently feasible**, which directly affects the 5% proposal criterion and later Methods/Experiments marks.

Reproducibility

"Fixed seeds" and a README are good, but insufficient. Add:

- exact parameter ranges;
- split ratios;
- preprocessing fitted only on index/training data;
- deterministic tie-breaking;
- package versions;
- hardware/threading policy;
- whether scaling subsets are nested;
- a configuration file or command for every reported experiment.

---

4. Experimental-design stress test

Splitting and leakage

Patient-grouped splitting is a strong choice because records are encounters and patient_nbr identifies patients. But specify: [archive.ics.uci.edu]

1. patient groups are split into **index-training, validation-query, and final test-query** sets;
2. ranges, category vocabularies, diagnosis grouping, pivot selection, and IVF centroids are fitted without test patients;
3. readmitted is excluded from all similarity features;
4. no query encounter is itself present in the index.

Repeated encounters inside the index may cause one patient to occupy several top-K positions. Report both encounter-level retrieval and a sensitivity analysis that retains at most one neighbor per patient.

Ties and duplicates

This dataset contains heavily discretized categorical features—age alone is provided in ten-year bins. Mixed 0/1 distances will therefore generate many tied distances. [archive.ics.uci.edu]

You need:

- deterministic object-ID tie-breaking;
- an exact duplicate count after preprocessing;
- a tie-aware recall sensitivity, or at minimum disclosure of how boundary ties at rank K are resolved.

Otherwise recall differences may measure tie ordering rather than retrieval quality.

Queries, seeds, and uncertainty

One thousand queries is reasonable operationally but is not automatically statistically sufficient. Report:

- per-query recall distributions;
- patient-level bootstrap confidence intervals;
- paired comparisons using the same queries across methods;
- variability over pivot seeds separately from variability over queries.

Mean ± SD over ten seeds alone hides query heterogeneity.

Parameter tuning

Do not run the full grid on the test set. Use validation patients to select a small Pareto set, freeze it, and evaluate those configurations once on test.

For fairness, compare at:

- matched index bytes;
- matched true-distance budget;
- matched total work;
- matched recall targets where possible.

Count all query work:

“Postings read” and “distance calculations” are not interchangeable operations, so neither alone supports a full efficiency claim.

Timing and memory

Use:

- single-threaded runs unless every method receives the same thread allowance;
- warm-up runs;
- repeated query batches;
- median and tail latency;
- build time separated from query time;
- actual serialized/index bytes;
- peak build memory and steady-state query memory separately.

---

5. Q3 and intrinsic dimensionality

Current Q3 cannot support its proposed conclusion. Increasing N does not manipulate intrinsic dimensionality; it mainly changes database size and possibly sample coverage.

You need either:

Option A — safer and recommended

Replace "intrinsic dimensionality" with measurable **distance concentration**:

- coefficient of variation of query-to-index distances;
- relative contrast between nearest-neighbor distance and mean/median distance;
- fraction of database distance evaluations performed by each exact tree;
- candidate fraction required for fixed recall.

Rewritten Q3:

> "As N increases, how do build cost, query work, and recall change, and how is pruning effectiveness associated with measured distance concentration under each representation?"

This supports evidence-based association without claiming a latent dimension.

Option B — retain intrinsic dimensionality

Preselect a named estimator, such as a local nearest-neighbor maximum-likelihood estimator, define its sampling protocol and uncertainty, and handle zero/tied distances explicitly. With this discrete dataset, tie-heavy distances can destabilize nearest-neighbor-ratio estimators, so include a tie-rate diagnostic.

Do not claim that intrinsic dimensionality "causes" pruning failure from a single observational dataset.

---

6. Dataset-specific threats

UCI reports 101,766 encounters, 47 current repository features, missing values, mixed categorical/integer data, patient and encounter IDs, ICD-related diagnoses, laboratory results, medications, and prior utilization. [archive.ics.uci.edu]

Key risks:

- **ICD-9 representation:** raw 0/1 mismatch treats every different code as equally dissimilar. Predefine clinically meaningful diagnostic groups or acknowledge the crude geometry.
- **Medication dominance:** 23 medication fields can overwhelm other concepts under equal per-column weighting. Use domain-group weights or report a grouped-weight sensitivity.
- **High-cardinality fields:** medical specialty and diagnosis increase sparsity and hurt trees.
- **Missingness semantics:** "test not performed" may be informative, while unknown race or payer is ordinary missingness. Do not apply one universal rule.
- **Label leakage:** readmitted and any derived readmission label must never enter distance construction.
- **Duplicate encounters:** duplicates or near-duplicates can inflate recall and create large tie sets.
- **Scaling claim:** 100k records supports a useful one-order-of-magnitude scaling study, but not a broad "massive-scale" claim. The MPT literature evaluates datasets with millions of objects, while the Jégou study validates at much larger vector scale. [link.springer.com], [irisa.fr]

Use "scaling within the available dataset," not "demonstrates large-scale scalability."

---

7. Computational feasibility

From the proposed N values:

Work item	Distance evaluations
Brute-force ground truth, 1 distance	185 million
Brute-force ground truth, 2 distances	370 million
Pivot build, $n = 128$, 10 seeds, 2 encodings	473.6 million
Pivot build, $n = 256$, 10 seeds, 2 encodings	947.2 million

These counts exclude feature-level operations, sorting, farthest-first construction, query pivot distances, postings accumulation, reranking, baselines, and the parameter grid.

This is feasible only with aggressive reuse and vectorized/compiled distance kernels. A custom Python loop over mixed features would be a major failure risk.

Scope cuts I strongly recommend

Keep:

- one primary mixed metric;
- brute-force ground truth;
- MIF and one precisely defined MPT;
- random versus farthest-first pivots;
- one exact tree baseline if the same metric can be represented correctly;
- $N = 10k, 50k, 100k$;
- $K = 10$, with $K = 50$ as sensitivity;
- validation-selected configurations and five to ten seeds.

Cut or defer:

- prefix index;
- prediction extension;
- a separate generic "PBI" method unless it is operationally distinct;
- one of KD-tree/ball-tree;
- IVF from the primary mixed-metric comparison;
- exhaustive Cartesian sweeps at every N.

Cache the largest pivot-distance table per pivot set and derive smaller $\tilde{n}$ and candidate budgets from it where methodologically valid.

---

8. Optional readmission prediction

As written, this extension is more likely to **dilute the project** than strengthen it. It adds:

- another tuning problem for K, voting, and class weights;
- prediction-specific leakage concerns;
- class imbalance;
- additional metrics and statistical comparisons;
- ambiguity over whether degradation arises from ANN search or weak kNN modeling.

Cut it from the proposal's promised work.

If core experiments finish, use it only as a tightly scoped downstream check:

> Fix one exact and two ANN operating points selected without prediction labels; compare whether replacing exact neighbors changes balanced accuracy and macro-F1 under an otherwise identical kNN rule.

No separate model search.

9. Prioritized revision list

A. Critical before submission

1. Replace the unconditional metric claim with the fixed-weight formulation and explicit missing-value rule.
2. Define MPT precisely and cite its direct source; separate it from permutation-prefix indexing.
3. State the truncated-permutation scoring and absent-pivot rule.
4. Restrict KD-tree/ball-tree to a metric and encoding they actually support.
5. Move IVF to the Euclidean sensitivity experiment or implement a genuinely comparable metric baseline.
6. Rewrite Q3 around distance concentration or define an intrinsic-dimensionality estimator.
7. Explicitly exclude labels and fit preprocessing only on training/index patients.
8. Remove the prediction and prefix-index extensions from committed scope.

B. Important improvements

- Add exact parameter ranges.
- Define tie handling and duplicate diagnostics.
- Compare methods at matched memory and work, not merely the same c.
- Use nested N-subsets and the same held-out queries where possible.
- Report actual bytes, full operation counts, confidence intervals, and per-query distributions.
- Moderate “large-scale” and clinical-impact language.

C. Optional polish

- Add a one-sentence threat-to-validity statement.
- Clarify what each team member will implement.
- Add small synthetic metric spaces with known answers as unit tests.
- Include a Pareto-front figure rather than large tables of every grid setting.

Suggested final research questions

Q1. *At fixed dataset size and metric, how do pivot count n, retained-prefix length $\tilde{n}$, and reranking budget c—including their interactions—affect recall@K, query work, and index bytes?*

Q2. *At matched index memory and candidate budget, does the precisely specified MPT representation achieve better recall–work trade-offs than truncated MIF, and does that conclusion depend on random versus farthest-first pivots?*

Q3. *As the index grows from 10k to 100k encounters, how do build cost, query work, and recall scale, and how is pruning effectiveness associated with measured distance concentration under the mixed and Euclidean representations?*

Overall assessment

Strongest aspects: excellent DS8001 topic fit; substantive implementation potential; sensible brute-force ground truth; patient grouping; validation/test separation; explicit computational metrics; randomized repetitions.

Biggest risks: an unjustified metricity claim, ambiguous MPT methodology, unsupported baseline equivalence, an unmeasured intrinsic-dimensionality claim, tie-heavy data, and a workload too broad for a two-person course project.

Highest-priority change: turn this from a broad benchmark into a controlled study of **MIF versus one precisely defined MPT under one defensible metric**, with brute force and one genuinely comparable baseline. That narrower version would likely be much easier to approve, execute correctly, and analyze at the depth the DS8001 rubric rewards.